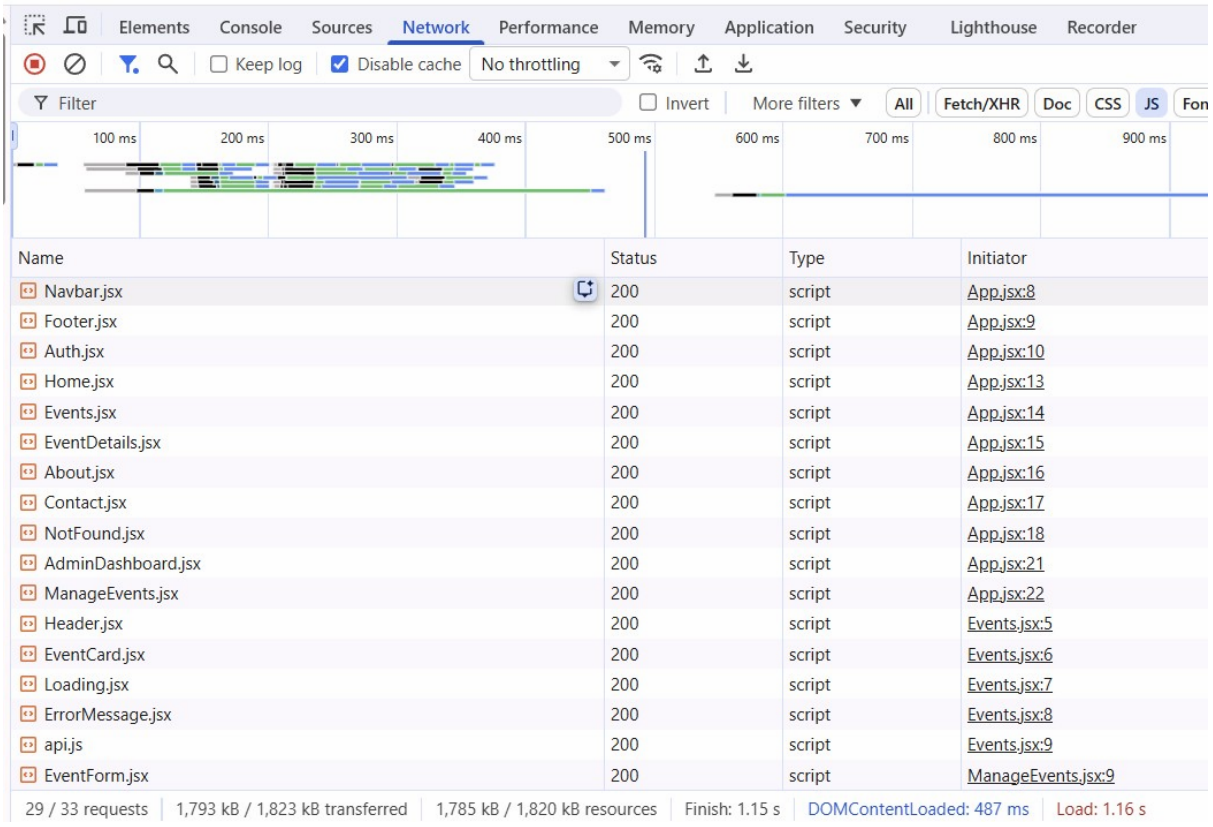
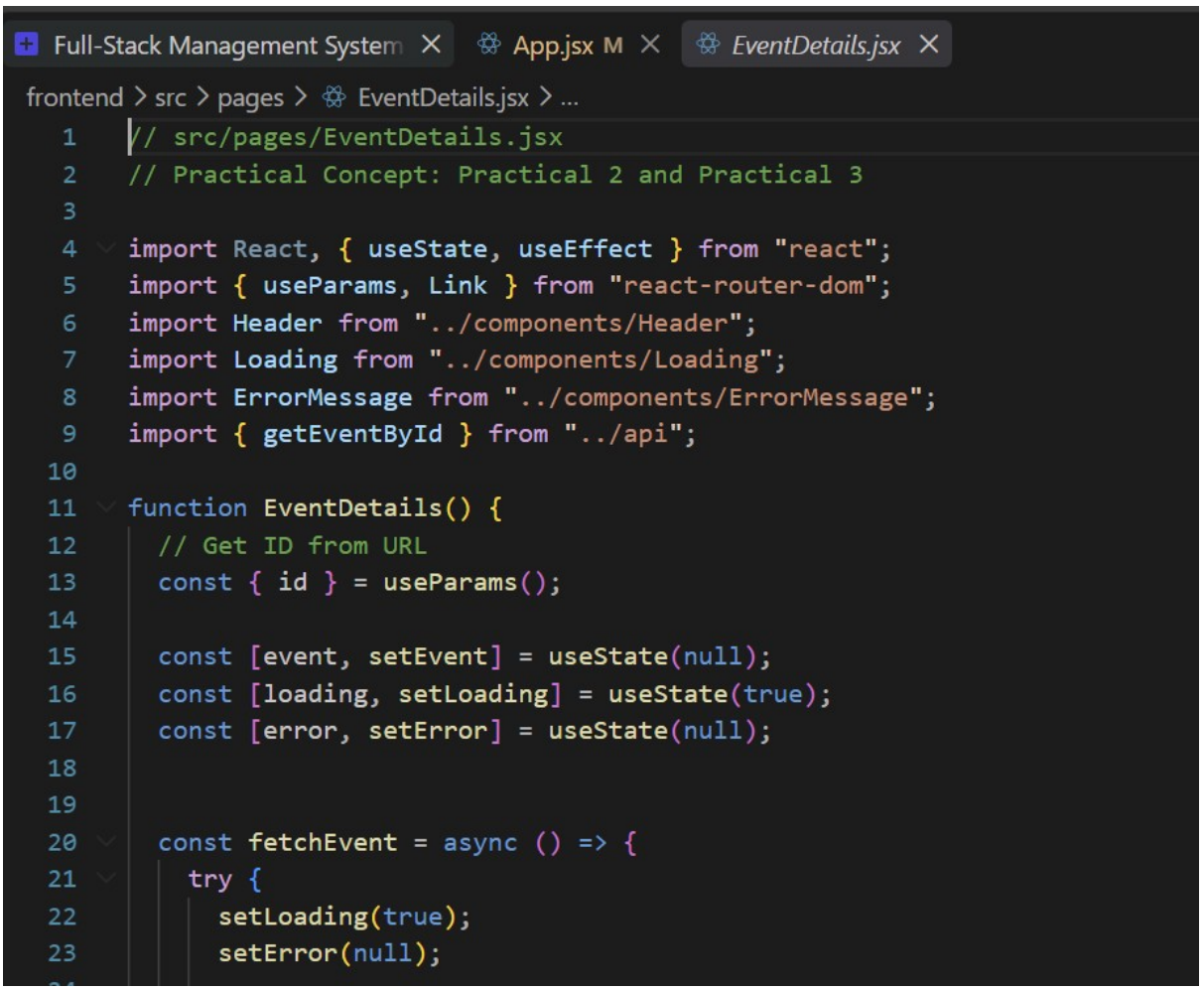
PRACTICAL-8

Aim: To improve the performance of a React application by implementing route-based lazy loading and code splitting using React.lazy() and Suspense, and to compare the application performance before and after optimization.



Modify App.jsx



Full-Stack Management System | App.jsx M | EventDetails.jsx

frontend > src > pages > EventDetails.jsx > ...

```jsx
// src/pages/EventDetails.jsx
// Practical Concept: Practical 2 and Practical 3

import React, { useState, useEffect } from "react";
import { useParams, Link } from "react-router-dom";
import Header from "../components/Header";
import Loading from "../components/Loading";
import ErrorMessage from "../components/ErrorMessage";
import { getEventById } from "../api";

function EventDetails() {
  // Get ID from URL
  const { id } = useParams();

  const [event, setEvent] = useState(null);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);


  const fetchEvent = async () => {
    try {
      setLoading(true);
      setError(null);
```

Full-Stack Management System | App.jsx M | EventDetails.jsx

frontend > src > pages > EventDetails.jsx > ...

```
      const response = await getEventById(id);

      setEvent(response.data);
    } catch (err) {
      setError(err.message || "Failed to load event.");
    } finally {
      setLoading(false);
    }
  };


  useEffect(() => {
    fetchEvent();
  }, [id]);


  if (loading) {
    return <Loading message="Loading event details..." />;
  }

  if (error) {
    return <ErrorMessage message={error} onRetry={fetchEvent} />;
  }
```

```
49    if (!event) {
50      return null;
51    }
52
53
54    return (
55      <div>
56        <Header
57          title={event.title}
58          subtitle="Event details"
59        />
60
61        <div className="form-card">
62          <h2>{event.title}</h2>
63
64          <p>
65            <strong>Description:</strong>
66            <br />
67            {event.description || "No description available."}
68          </p>
69
70          <p>
71            <strong>Event Date:</strong>{" "}
72            {new Date(event.eventDate).toLocaleDateString()}
```

frontend > src > pages > EventDetails.jsx > ...

```jsx
            {new Date(event.eventDate).toLocaleDateString()}
          </p>

          <p>
            <strong>Location:</strong> {event.location}
          </p>

          <p>
            <strong>Category:</strong> {event.category}
          </p>

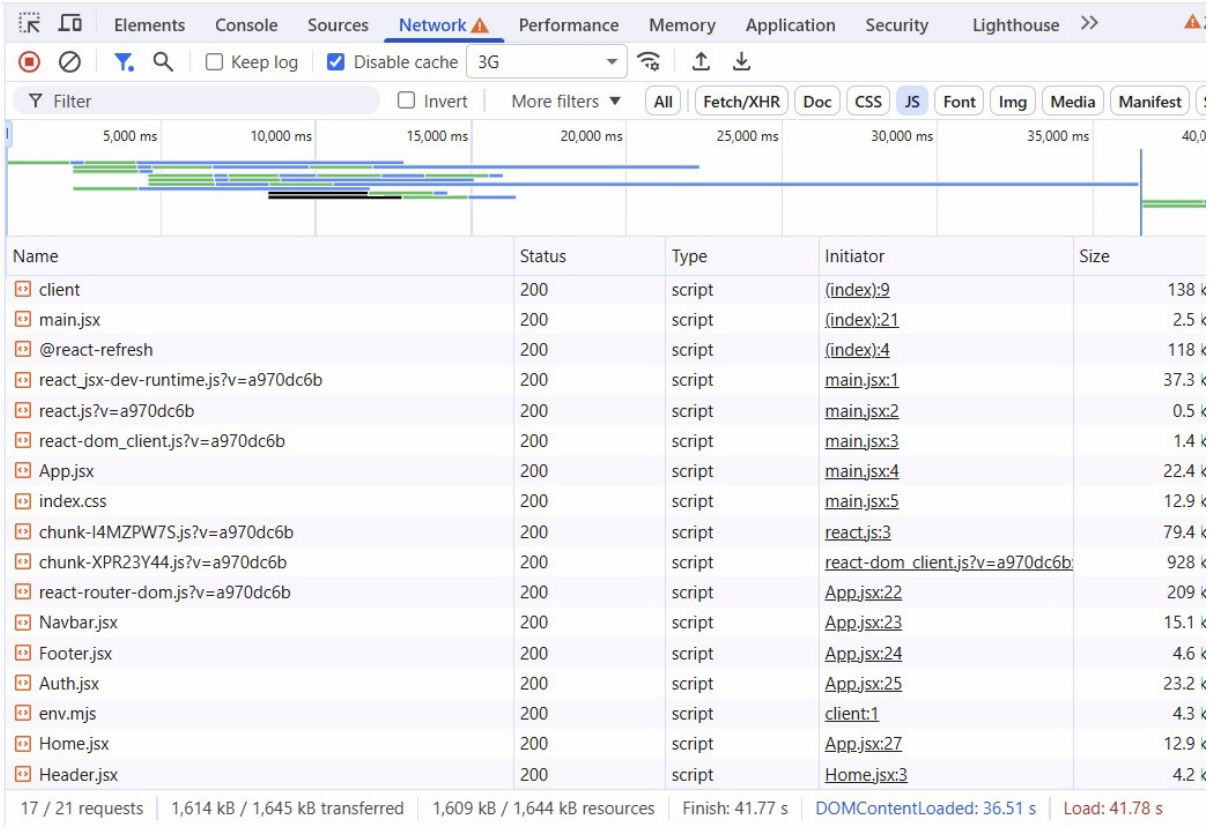
          <p>
            <strong>Status:</strong> {event.status}
          </p>

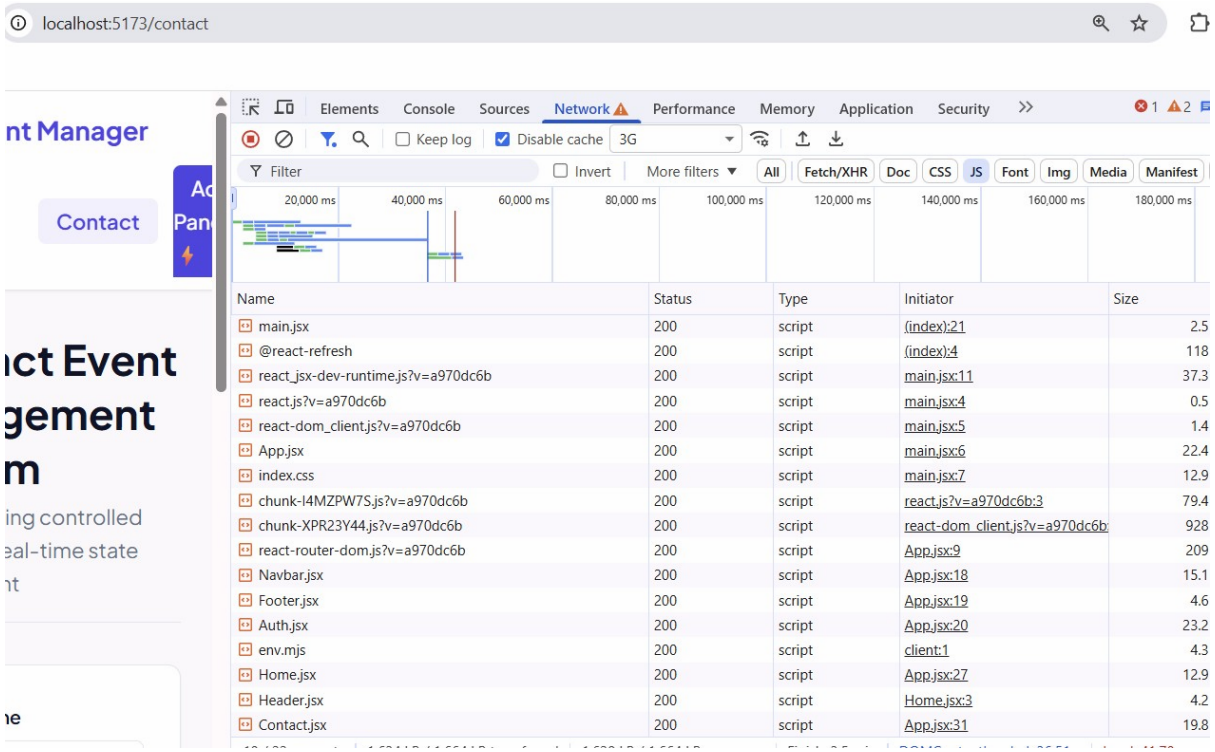
          <Link to="/events" className="btn btn-secondary">
            ← Back to Events
          </Link>
        </div>
      </div>
    );
  }

  export default EventDetails;
```

```
> full-stack-management-frontend@1.0.0 build
> vite build

vite v5.4.21 building for production...
✓ 52 modules transformed.
dist/index.html                          0.96 kB │ gzip:  0.57 kB
dist/assets/index-B4GtJEew.css           7.49 kB │ gzip:  2.05 kB
dist/assets/Header-DOrFi-tr.js           0.20 kB │ gzip:  0.17 kB
dist/assets/NotFound-CKnJ9AfB.js         0.52 kB │ gzip:  0.33 kB
dist/assets/EventDetails-D2rF8_cU.js     1.22 kB │ gzip:  0.58 kB
dist/assets/Home-Ca9cIcQa.js             1.52 kB │ gzip:  0.63 kB
dist/assets/About-YijY2DcY.js            1.56 kB │ gzip:  0.72 kB
dist/assets/Events-nOkkfh1E.js           1.67 kB │ gzip:  0.82 kB
dist/assets/api-CFir29gr.js              1.99 kB │ gzip:  0.74 kB
dist/assets/Contact-CuYMX3mu.js          2.21 kB │ gzip:  0.97 kB
dist/assets/AdminDashboard-BWUVTohu.js   2.37 kB │ gzip:  0.95 kB
dist/assets/ManageEvents-BKzivHIm.js     5.71 kB │ gzip:  1.89 kB
dist/assets/index-Duv4wOzO.js          172.09 kB │ gzip: 56.34 kB
✓ built in 2.04s
```


Elements Console Sources Network Performance Memory Application Security Lighthouse
Keep log Disable cache 3G
Filter Invert More filters All Fetch/XHR Doc CSS JS Font Img Media Manifest
5,000 ms 10,000 ms 15,000 ms 20,000 ms 25,000 ms 30,000 ms 35,000 ms
Name Status Type Initiator Size
client 200 script (index):9 138 k
main.jsx 200 script (index):21 2.5 k
@react-refresh 200 script (index):4 118 k
react_jsx-dev-runtime.js?v=a970dc6b 200 script main.jsx:1 37.3 k
react.js?v=a970dc6b 200 script main.jsx:2 0.5 k
react-dom_client.js?v=a970dc6b 200 script main.jsx:3 1.4 k
App.jsx 200 script main.jsx:4 22.4 k
index.css 200 script main.jsx:5 12.9 k
chunk-I4MZPW7S.js?v=a970dc6b 200 script react.js:3 79.4 k
chunk-XPR23Y44.js?v=a970dc6b 200 script react-dom_client.js?v=a970dc6b 928 k
react-router-dom.js?v=a970dc6b 200 script App.jsx:22 209 k
Navbar.jsx 200 script App.jsx:23 15.1 k
Footer.jsx 200 script App.jsx:24 4.6 k
Auth.jsx 200 script App.jsx:25 23.2 k
env.mjs 200 script client:1 4.3 k
Home.jsx 200 script App.jsx:27 12.9 k
Header.jsx 200 script Home.jsx:3 4.2 k
17 / 21 requests 1,614 kB / 1,645 kB transferred 1,609 kB / 1,644 kB resources Finish: 41.77 s DOMContentLoaded: 36.51 s Load: 41.78 s


localhost:5173/contact
nt Manager
Contact
ict Event
gement
m
ing controlled
eal-time state
Elements Console Sources Network Performance Memory Application Security
Keep log Disable cache 3G
Filter Invert More filters All Fetch/XHR Doc CSS JS Font Img Media Manifest
20,000 ms 40,000 ms 60,000 ms 80,000 ms 100,000 ms 120,000 ms 140,000 ms 160,000 ms 180,000 ms
Name Status Type Initiator Size
main.jsx 200 script (index):21 2.5
@react-refresh 200 script (index):4 118
react_jsx-dev-runtime.js?v=a970dc6b 200 script main.jsx:11 37.3
react.js?v=a970dc6b 200 script main.jsx:4 0.5
react-dom_client.js?v=a970dc6b 200 script main.jsx:5 1.4
App.jsx 200 script main.jsx:6 22.4
index.css 200 script main.jsx:7 12.9
chunk-I4MZPW7S.js?v=a970dc6b 200 script react.js?v=a970dc6b:3 79.4
chunk-XPR23Y44.js?v=a970dc6b 200 script react-dom_client.js?v=a970dc6b 928
react-router-dom.js?v=a970dc6b 200 script App.jsx:9 209
Navbar.jsx 200 script App.jsx:18 15.1
Footer.jsx 200 script App.jsx:19 4.6
Auth.jsx 200 script App.jsx:20 23.2
env.mjs 200 script client:1 4.3
Home.jsx 200 script App.jsx:27 12.9
Header.jsx 200 script Home.jsx:3 4.2
Contact.jsx 200 script App.jsx:31 19.8

Github: https://github.com/sahilmak1602/WDF_24IT045_SAHIL.git